

3 (a) (i) $v_{\max} = 1.4 \text{ m s}^{-1}$

$$E_K = \frac{1}{2} m v_{\max}^2$$

$$= \frac{1}{2} (0.50) (1.4)^2 \quad [\text{C1}]$$

$$= 0.49 \text{ J} \quad [\text{A1}]$$

(ii) $v_{\max} = \omega x_0$

$$x_0 = \frac{1.4}{\left(\frac{2\pi}{0.50} \right)} \quad [\text{C1}]$$

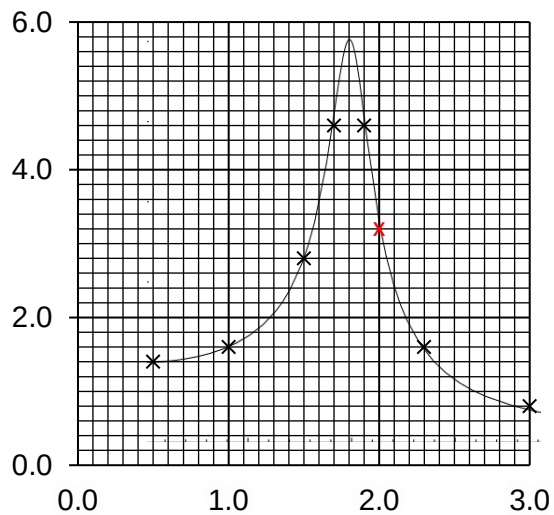
$$= 0.11 \text{ m} \quad [\text{A1}]$$

(b) Damped oscillation. [B1]

Air resistance [B1] on the wooden board.

(c) (i)

(ii)



Correct plot (2.0, 3.2) [B1]

Correct curve of best fit [B1]

(iii) 1.8 Hz (accept 1.75, 1.8, 1.85) [B1]